



Department of Electronics and Communication Engineering

Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. ECE A

Course Code & Title: EC8691 Microprocessor and Microcontroller

Name of the Faculty member: Mr.D.Gopinath

Innovative Practice Description

- **Unit / Topic:** Unit III / Case studies - LCD Display, Keyboard Display Interface and Alarm Controller
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO10
- **Activity Chosen:** Flipped Classroom
- **Justification:**

A flipped classroom is a type of blended learning where students are introduced to content at home and practice working through it at class room. Here, case studies of LCD Display, Keyboard Display Interface and Alarm Controller was given to three different student groups. In that, different interfacing components with 8086 microprocessor can be learned as a group. In a group, students are worked in collaborative manner and present their case study in front of other groups.

- **Time Allotted for the Activity:** One Period (50 Minutes)

- **Details of the Implementation:**

At first the students are divided into groups of 13 or 14 members (3 Groups). Discussed the Flipped Classroom activity to the students. Asked the preparedness of each group about the case studies

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PSO2
CO3	3	3	2	2	2	3	3

- **PO / PSO mapped:**

Innovative practice	PO2	PO8	PO9	PO10	PSO2
	3	2	2	3	3
Justification for correlation	The course outcome is highly	Identify tenets of the professional	A collaborative activity	The topics deals to different	The different interfacing components

	correlated since it involves the analysis of different interfacing components with 8086 microprocessor	code of ethics while doing activity-based learning. So, it is moderately correlated	Flipped classroom is planned which involves Implement the norms of practice (e.g. rules, roles, charters, agendas, etc.) of effective team work, to accomplish a goal. This course outcome is correlated moderately.	interfacing components with 8086 microprocessor to complement writing and presentations. This task is accomplished through assignment therefore course outcome is correlated highly.	with 8086 microprocessor are required for design of embedded systems. So, the course outcome is highly correlated.
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• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

Students feel that it was interesting. They learn how to communicate with team members and work together.

❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
Students can learn at their own pace. They take responsibility for their learning. They learn rather than encounter material in class. Because of this activity, increased collaboration between students.

❖ ***Challenges faced in implementation:***

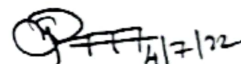
Two student groups did the presentations effectively. However, one of the student groups had different thoughts while presenting the topic and also they have not communicated effectively with in that group.

References:

- ❖ Yu-Cheng Liu, Glenn A.Gibson, "Microcomputer Systems: The 8086 / 8088 Family - Architecture, Programming and Design", Second Edition, Prentice Hall of India, 2007.
- ❖ <https://www.teachthought.com/learning/definition-flipped-classroom/>
- ❖ <https://bokcenter.harvard.edu/flipped-classrooms>



Signature of Faculty Member



HOD



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Active Learning Activity- Unit-I- The 8086 Microprocessor

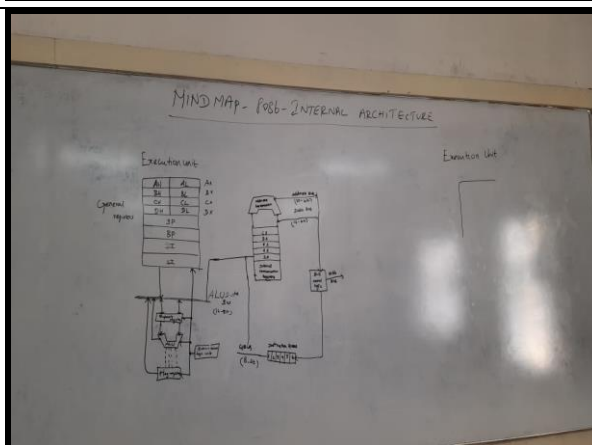
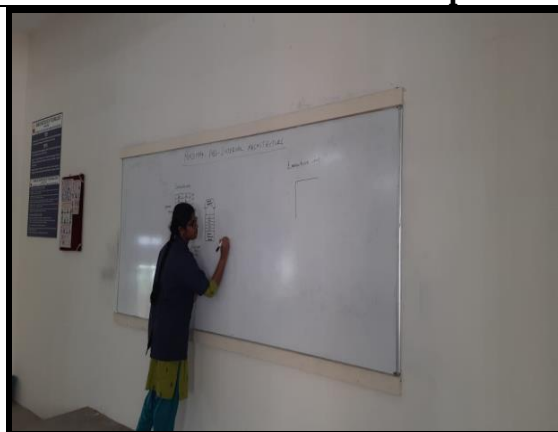
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Name of the Topic	Name of the Activity	Date & Duration
8086 Microprocessor Architecture	Mind map	11.03.2022 & 5 Minutes
Byte and String Manipulation and Revision	One Minute Paper	28.03.2022 & 5 Minutes

8086 Microprocessor Architecture - Mind map

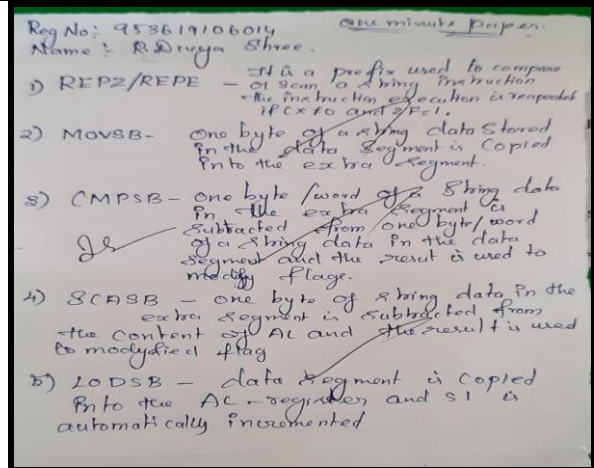
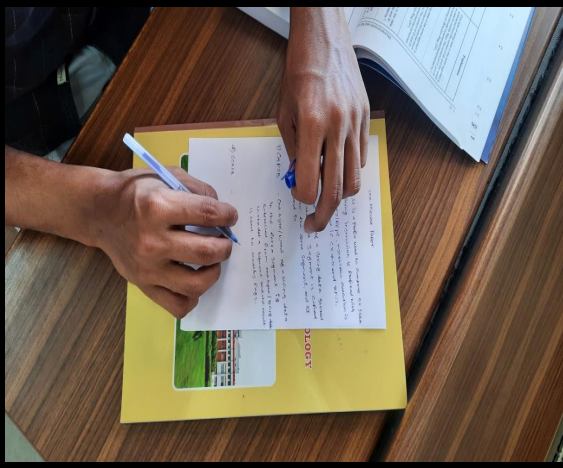




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Byte and String Manipulation and Revision - One Minute Paper





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Active Learning Activity- Unit-II- 8086 System Bus Structure

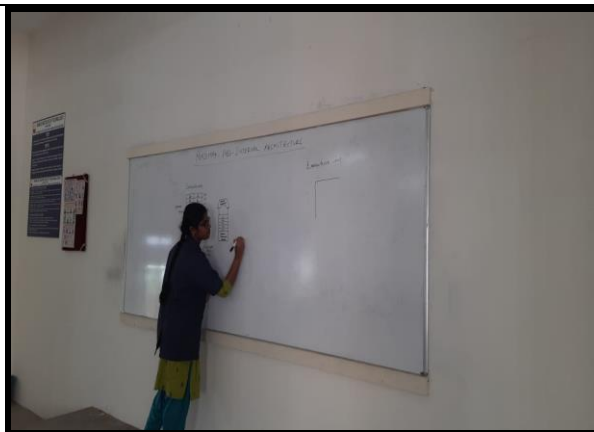
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Name of the Topic	Name of the Activity	Date & Duration
System Design using 8086	Z-A Approach	5.4.2022 & 5 Minutes
Closely Coupled and Loosely Coupled Configurations, Introduction to Advanced Processors - 80286, 80386 & 80486 and Revision	Exit Slip	19.4.2022 & 5 Minutes

System Design using 8086- Z-A Approach

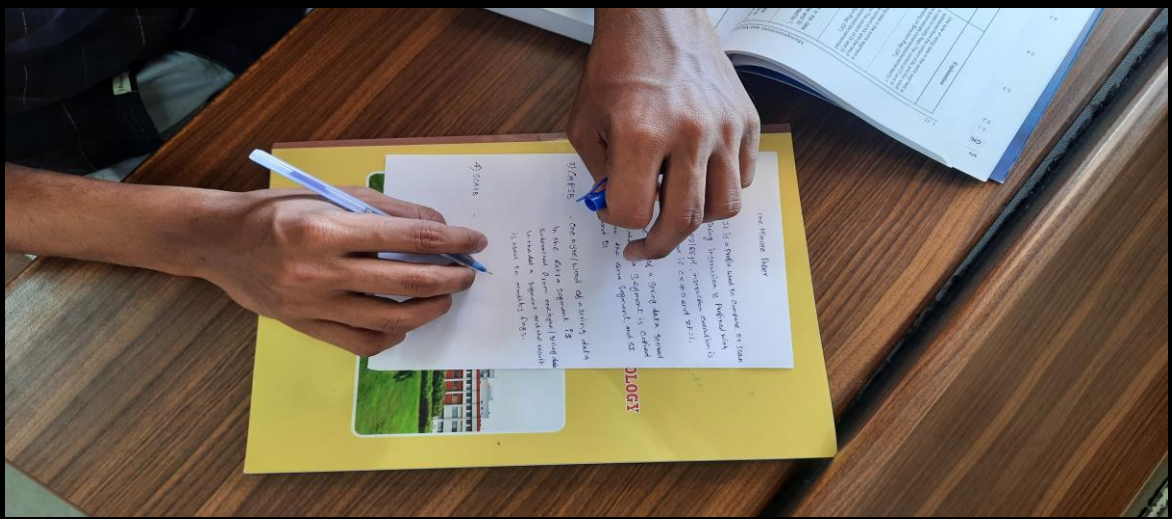
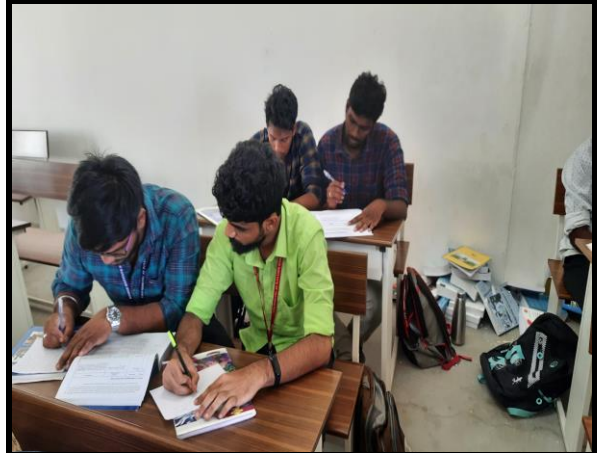




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Closely Coupled and Loosely Coupled Configurations, Introduction to Advanced Processors - 80286, 80386 & 80486 - Exit Slip





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Active Learning Activity- Unit-III- I/O INTERFACING

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Name of the Topic	Name of the Activity	Date & Duration
D/A interface and A/D Interface	Brain Storming	10.5.2022 & 5 Minutes
Case studies - LCD Display, Case studies - Keyboard Display Interface and Alarm Controller	Flipped Classroom	17.5.2022 & 50 Minutes

D/A interface and A/D Interface - Brain Storming





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Case studies - LCD Display, Case studies - Keyboard Display Interface and Alarm Controller- Flipped Classroom



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